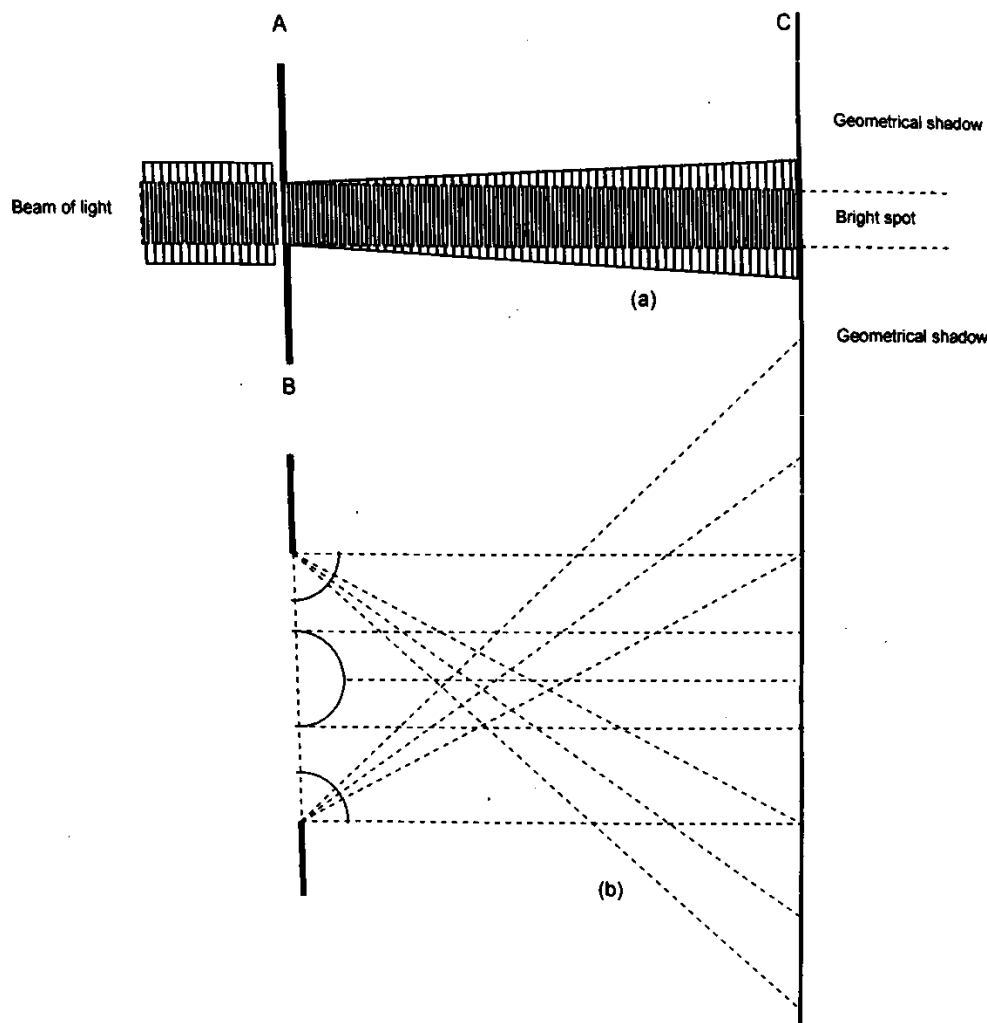


Diffraction of Light:

"When a beam of light passes through a narrow slit, it spreads out to a certain extent into the geometrical shadow. This phenomenon is called diffraction". In other words, "the bending of light round corners and spreading of light into the geometrical shadow is called diffraction." This bending of light that is responsible for encroachment of light beam into the geometrical shadow, is consistent with the spreading of Huygens' construction. However, diffraction occurs for waves of all types, not just for light wave.



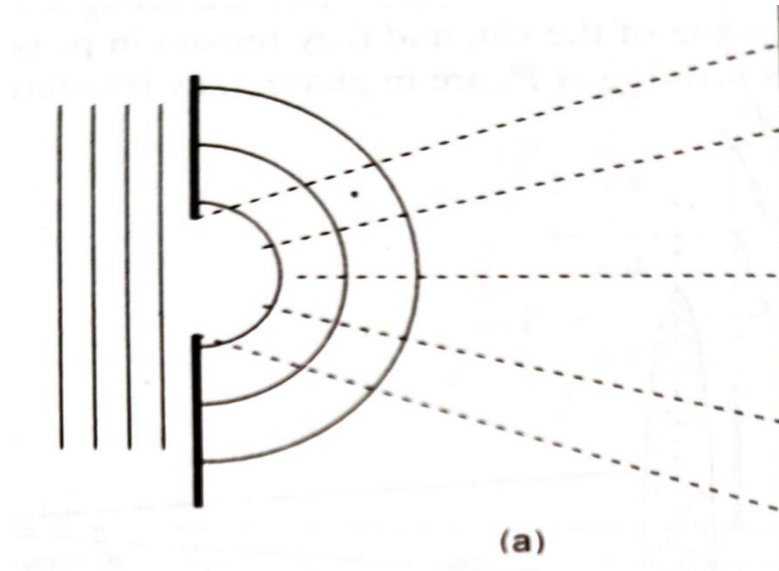
In order to explain the phenomenon, let us consider a narrow beam of light is falling on to the slit of the barrier AB, where the beam width is smaller than that the slit width (that means the wavelength is smaller than the slit width) then we will see a bright spot on the screen C of the same size of the slit because light travels in straight line in the same medium according to the geometric optics but if the beam width is larger than the slit width (that means the wavelength is larger than that of the slit width), then we see that the bright spot is larger in size than that of the slit width, which is caused by the bending of light at the edges of the slit and results in encroachment into the geometric shadow as shown in figure (a). This bending of light is caused because all the points in the plane of the slit will act as secondary sources of light according to the Huygens' construction as shown in figure (b). Experimentally it is found that the narrower the slit the greater the diffraction.

Classification of Diffraction:

Diffraction are of two types, viz-

1. Fresnel class of diffraction
2. Fraunhofer class of diffraction

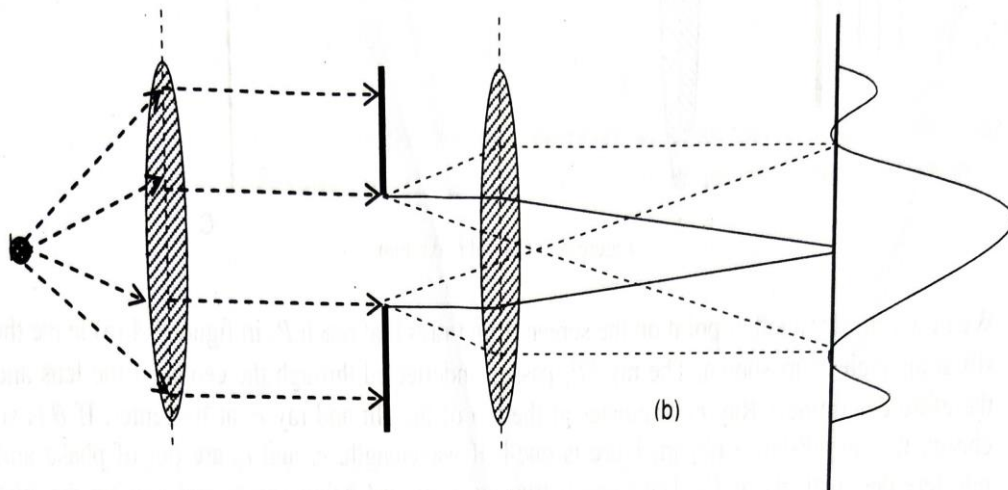
1. **Fresnel class of Diffraction:** In the Fresnel class of diffraction, the source or the screen or the both are at finite distance from the aperture 'or obstacle causing diffraction.



In this case the incident wave front is either spherical or cylindrical.

This type of diffraction is found when spherical wave front falls 'on a needle, thin wire or slit.

2. **Fraunhofer class of diffraction:** In the Fraunhofer class of diffraction, the source and the screen both are at infinite distance from the aperture or the obstacle causing diffraction.



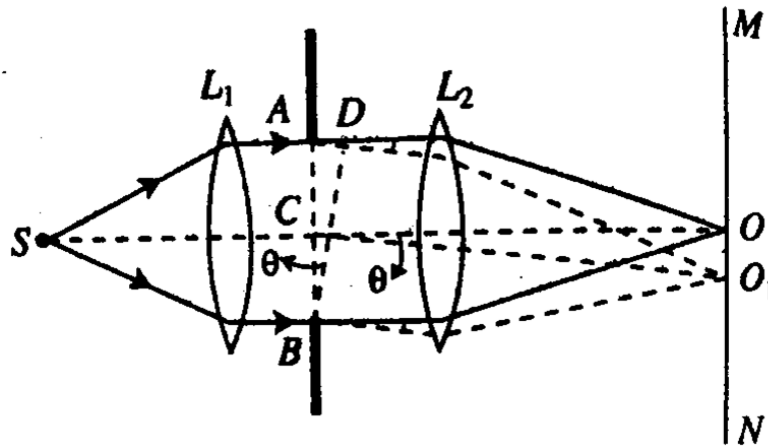
Plane wave front causes this type of diffraction.

If a source of light is placed at the focal plane of a convex lens then after refraction through the lens incident on an obstacle or aperture then Fraunhofer class of diffraction is obtained.

Single or double slit, diffraction grating etc produce Fraunhofer class of diffraction.

Diffraction at a Single Slit:

Let us consider narrow slit AB. The width of this slit is a . The slit is kept perpendicularly to the plane of the book. A convex lens L_1 is placed in front of the slit. Monochromatic light of wavelength λ emitted from the slit S is placed at the principal focus of the convex lens become parallel rays of light after refracting through the lens [Figure]. These parallel rays in the form of plane wave fronts fall on the slit AB. With the help of another lens L_2 the rays of light emitted from AB are centralized on the screen MN kept at the focal plane of L_2 .



While going through the slit AB the rays of light also spread above the point A and also below the point B. So instead of producing a sharp image of the slit at O alternate bright and dark diffraction fringes are formed on either side of O. All the particles on the plane wave front situated at the slit AB are in phase. As a result, secondary waves from those particles travel parallel to the axis of the slit CO and after refracting through the lens L_2 converges to the point O. In this way the rays of light reach at O with same phase and produce constructive interference and the brightness of that point becomes maximum. The point O is called the principal maximum.

Now let us consider that few secondary waves are diffracted making an angle with CO and incident on the lens L_2 going parallel to CO_1 . Now these rays converge to O_1 after being refracted through the lens L_2 . In this case the waves do not travel equal path so their phase at O_1 will not be same. Whether the intensity at O_1 will be maximum or minimum that depends on the path difference of the secondary waves.

From figure it is seen that, the path difference of the two secondary waves emitted from the two ends A and B of the slit AB is,

$$AD = AB \sin \theta = a \sin \theta$$

The conditions that the point O_1 will be maximum or minimum can be found by mathematical analysis.

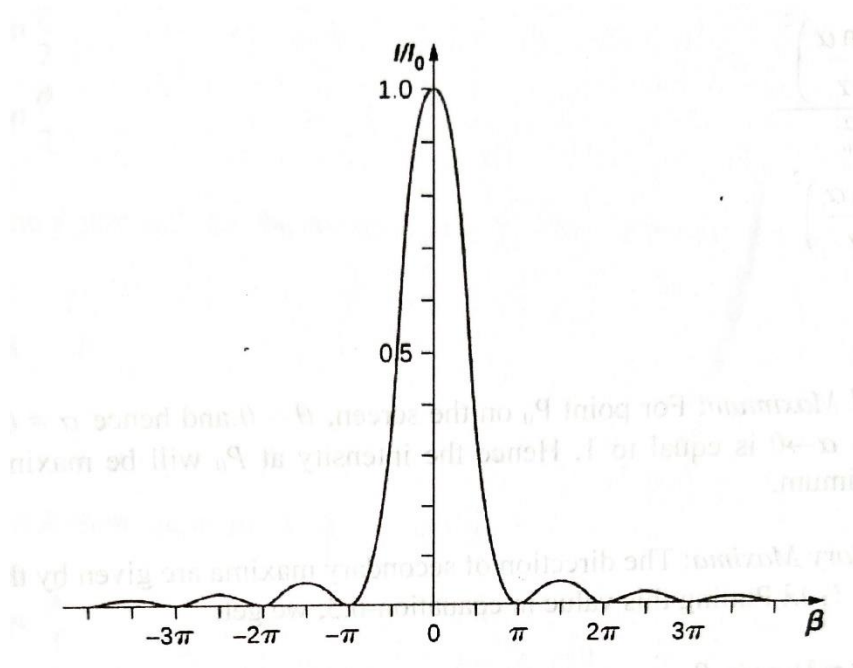
Condition for Minimum: if θ_n be the angle of diffraction for the n^{th} minimum on either sides of the point O, then –

$$a \sin \theta_n = n\lambda, \quad \text{Where, } n = 1, 2, 3, \dots \dots \dots$$

Condition for Maximum: if θ_n be the angle of diffraction for the n^{th} maximum of either sides of the point O, then –

$$a \sin \theta_n = (2n + 1) \frac{\lambda}{2},$$

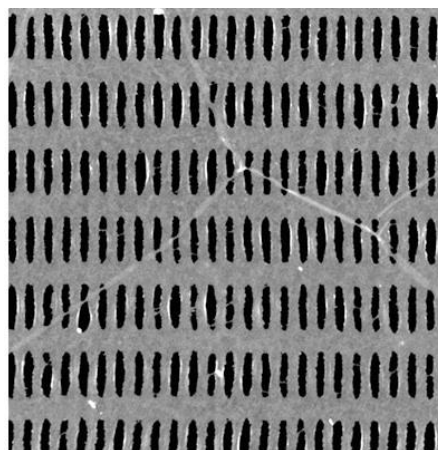
Intensity Distribution of Diffraction Pattern:



Diffraction Grating:

Diffraction grating is an important device to analyze light source. It consists of large number of parallel slits of equal width.

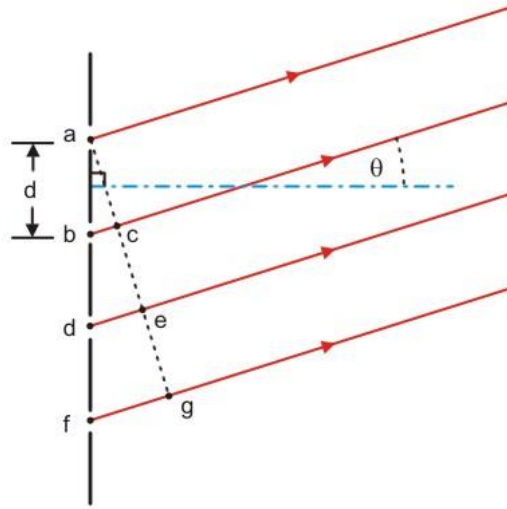
Definition: A sheet having a large number of narrow slits of equal width lying side by side is called a grating.



Method of construction: Gratings are prepared by ruling equidistant parallel lines on a glass surface. The lines are drawn with a fine diamond point. The space between any two lines is transparent to light and the lined portion is opaque. The gratings, which we use in our laboratory, contain about 10,000 lines per centimeter. As a result, the width of a slit is about 10^{-4} cm.

Grating Constant or Grating Element: The distance between the beginning of one slit to the beginning of the next slit is called the grating constant.

The width of one transparent portion and the width of one opaque portion of a grating is the grating constant.



Let, width of each slit = a

width of each line = b

Grating constant, $d = a + b$

So, in a grating the number of lines in distance d is one. Therefore, in unit length the number of lines is $\frac{1}{d}$. Now if N is the number of lines per unit length of a grating then,

$$N = \frac{1}{d}$$

$$\text{Or, } N = \frac{1}{a+b}$$

After analyzing the spectra produced by diffraction grating, it has been shown that the position of the principal maxima is given by –

$$d \sin \theta = n\lambda \quad \text{Where, } n = 0, 1, 2, 3, \dots$$

$$\text{Or, } (a + b) \sin \theta = n\lambda$$

$$\text{Or, } \sin \theta = \frac{n\lambda}{(a+b)}$$

This relation is also known as grating equation and can be used to study the dependence of the angle of diffraction θ on the wavelength λ .

Resolving Power of Grating:

Resolving Power: The resolving power of a grating is a measure of its ability to spatially separate two wavelengths. It is determined by applying the Rayleigh criteria to the diffraction maxima; two



wavelengths are resolvable when the maxima of one wavelength coincides with the minima of the second wavelength.

The chromatic resolving power (R) is defined by –

$$R = \frac{\lambda}{\Delta\lambda} = nN$$

where $\Delta\lambda$ is the resolvable wavelength difference, n is the diffraction order, and N is the number of lines of grating per unit centimeter.

Dispersive Power of Grating:

The dispersive power of a grating is defined the ratio of the difference in angle of diffraction of any two neighbouring spectral lines to the diffraction in the wavelength between the two spectral lines. It is expressed as –

$$\frac{d\theta}{d\lambda} = \frac{n}{d \cos \theta}$$

Problem-1: How many orders will be visible (Max. observable angle is 30°) if the wavelength of incident radiation be $5000\text{\AA}$. and the number of lines on the grating be 14000 per inch? [Answer: 3.63]

Problem-2: The limits of the visible spectrum are approximately $4000\text{\AA}$. and $7000\text{\AA}$. Find the angular breadth of the first order visible spectrum produced by a plane diffraction grating having 15000 lines per inch when light is incident normally on the grating. [Answer: $10^\circ 50'$]

Problem-3: A diffraction grating having 4000 lines to a cm is used at normal incidence. Calculate the dispersive power of the grating in the third order spectrum in the wavelength region $5000\text{\AA}$. [Answer: 15000]

Problem-4: Calculate the angle between the lamp filament and its first diffracted image produced by a fabric with 160 threads/cm. ($\lambda = 6000\text{\AA}$.) [Answer: 33min]

Problem-5: A diffraction grating used at normal incidence gives a green line ($\lambda = 5400\text{\AA}$.) in a certain order superimposed on the violet line ($\lambda = 4050\text{\AA}$.) of the next higher order. If the angle of diffraction is 30° , how many lines are there to the centimeter in the grating? [Answer: 3086 cm^{-1}]

Problem-6: Calculate the minimum number of lines in a grating which just resolve the sodium lines in the first order spectrum. The wavelengths are $5890\text{\AA}$. and $5896\text{\AA}$. [Answer: 982 appr]x]